

Chemistry Ch 5.4: Intermolecular Forces

Chapter 5 · kitab ke page 73 se 94

Ek sawal. Paani sau digree par ubalta hai. Lekin H_2S , jo paani se **bhaari** hai, manfi saath digree par hi gas ban jati hai. Bhaari cheez pehle ubal gayi? Ye ulta kyun hai?

Pehle ek farq jo sab kuch khol deta hai

Intramolecular force: molecule ke **andar**, atoms ke darmiyan. Yehi ionic aur covalent bond hain, aur ye **mazboot** hain.

Intermolecular force: molecule ke **bahar**, ek molecule aur doosre molecule ke darmiyan. Ye **kamzor** hain.

Aur ubalne ka taalluq **intermolecular** se hai, kyunke ubalte waqt molecules ek doosre se alag hote hain, tootte nahin.

Do qism ki intermolecular forces

1. Dipole dipole forces: polar molecules mein ek sira halka musbat aur doosra halka manfi hota hai. Ek molecule ka musbat sira doosre ke manfi sire ko **kheenchna** hai.

2. Hydrogen bonding: ye dipole dipole ki **sab se mazboot** shakal hai. Ye tab banti hai jab hydrogen **N, O ya F** ke saath jura ho.

Teen harf jo yaad rakhne hain: **N, O, F**

Hydrogen bonding sirf in **teen** ke saath banti hai. Kyun? Kyunke ye teenon **bohat chhote** aur **bohat electronegative** hain, to wo hydrogen ka electron poori tarah apni taraf kheench lete hain aur hydrogen taqreeban nanga proton reh jata hai.

Aur yehi oopar wale sawal ka jawab hai: paani mein hydrogen bonding hai, H_2S mein nahin (kyunke S in teen mein nahin hai). Isi liye paani ko alag karne ke liye zyada garmi chahiye.

Quiz

1. Explain the difference between **intramolecular** and **intermolecular** forces, and say which of them decides a substance's boiling point.
2. Name the two intermolecular forces your book gives. For **hydrogen bonding**, name the three elements hydrogen must be bonded to, and use it to explain why water boils at a much higher temperature than H_2S .